



A survey of real-time surface defect inspection methods based on deep learning

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Abstract

In recent years, deep learning methods have been widely used in various industrial scenarios, promoting industrial intelligence. Real-time surface defect inspection of industrial products is one of the research focuses in industry. Surface defect inspection methods based on deep learning show great advantages and make it possible to detect defects in real time with high accuracy. From the perspective of real-time inspection, according to different types of surfaces in industry, this paper reviews the latest deep learning-based surface defect inspection methods from three levels: defect classification, defect detection and defect segmentation. After that, this paper introduces commonly used metrics for evaluating the performance of surface defect inspection models and public surface defect datasets. Then, this paper discusses the challenges faced by deep learning-based real-time surface defect inspection methods, including the acquisition of surface defect datasets, balancing the accuracy and speed of inspection models, and the application in industrial environments. Finally, this paper provides an outlook on the future development trend of surface defect inspection.

Keywords Deep learning · Computer vision · Surface defect inspection · Real-time inspection

1 Introduction

During industrial production, the inspection of product surface defects is an indispensable process, and is the key to product quality control. Product quality affects the efficiency of production, and early inspection of product defects can reduce the waste of raw materials and reduce production costs. In addition, product quality is related to the reputation and market share of the factory. How to quickly and accurately detect product surface defects has become one of the important research focuses in industry.

Surface defect inspection in industry is initially carried out by manual inspection. Due to the limited energy and attention of human beings, product surface defect inspection not only has low accuracy, but also takes a long time, which increases labor costs and seriously

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hinders the improvement of production efficiency. With the development of machine learning and artificial intelligence, image-based automatic surface defect inspection is widely used in various industrial scenarios and replaces manual inspection. Current surface defect inspection methods can be mainly divided into traditional methods (manual feature extraction) and deep learning-based methods (automatic feature extraction). Traditional methods extract defect features through image processing, image analysis, etc., while deep learning-based methods learn defect features through a certain number of defect samples, and then automatically extract defect features.

Traditional methods can be divided into three categories: statistical methods, spectral methods and model-based methods. Statistical methods count and evaluate the pixels of defect images, and analyze the spatial distribution of pixels to detect surface defects. Chu et al. (2017) proposed a method to classify steel surface defects based on various types of statistical features. These statistical features are statistical descriptions of one-dimensional and two-dimensional histograms of local binary pattern (LBP) operators. Aminzadeh and Kurfess (2015) selected an appropriate threshold for surface defect inspection by comparing the histograms of the background and defect areas. Spectral methods transform the defect image in the space domain, frequency domain, etc., and realize surface defect inspection according to the different responses of the defect-free area and the defect area. Bai et al. (2014) obtained the saliency area of the defect image according to the phase-only Fourier transform, and then compared the saliency area with the corresponding area of the defect-free template to detect surface defects. Yang et al. (2016) proposed a method to automatically detect surface defects of magnetic tiles under various lighting conditions based on Stationary Wavelet Transform. Model-based methods map the original texture distribution of defect images to a low-dimensional distribution to better detect various surface defects. Liu et al. (2017) proposed the Haar-Weibull-Variance model for steel surface defect inspection. The Haar-Weibull-Variance model only needs to learn two parameters to map the texture distribution of the image patch to a low-dimensional spatial distribution. Cao et al. (2017) proposed a subspace segmentation model based on least squares regression, and combined local prior knowledge with the model to achieve the segmentation of fabric surface defects. Although traditional methods have achieved certain applications in industry, there are two common problems with traditional methods: one is that the features need to be designed manually, this process requires expert knowledge, and the quality of the features directly affects the performance of surface defect inspection; the other is the inaccurate defect location, only the rough location of the defect can be obtained. In addition, the complex industrial environment makes the acquired images have low contrast, noise, etc., making it difficult for traditional methods to achieve satisfactory inspection performance.

In recent years, with the development of deep learning, deep learning-based methods have been widely used in various industrial scenarios. With powerful feature extraction capabilities, convolutional neural network (CNN) has shown strong performance in surface defect inspection of industrial products. Compared with traditional methods, deep learning-based methods can automatically extract defect features, and the deep learning model is robust, enabling high-accuracy inspection even in complex industrial environments. Currently, surface defect inspection methods based on deep learning are widely used in steel strip (He et al. 2019b; Di et al. 2019; Song et al. 2021), welding (Sassi et al. 2019; Le et al. 2020), aluminum profile (Wei and Bi 2019; Chen et al. 2021a; Zeng et al. 2021), solar cell (Bartler et al. 2018; Su et al. 2020), fabric (Zeng et al. 2021; Chen et al. 2022) etc., and have achieved satisfactory inspection performance. From several industry-related journals, this paper counts the number of surface defect inspection papers and the number of

real-time surface defect inspection papers in the past 5 years, as shown in Fig. 1. In recent years, both the number of surface defect inspection papers and the number of real-time surface defect inspection papers have been increasing. Therefore, it is of great significance to review related research on surface defect inspection and real-time surface defect inspection. Bhatt et al. (2021) presented surface defect inspection from three different perspectives, including defect detection context, learning techniques, and defect localization and classification. Tulbure et al. (2022) reviewed object detection-based surface defect inspection methods and briefly summarized model compression and acceleration techniques. Yang et al. (2020b) firstly classified the defects of products, and secondly, reviewed the latest mainstream surface defect inspection techniques, and analyzed the application situation of the major defect inspection equipment. They provided reference for defect inspection technology in theory and practical application. In addition, there are many reviews of surface defect inspection methods for a certain surface. Li et al. (2021) introduced traditional and deep learning-based methods for fabric surface defect inspection, and compared these deep learning-based methods. Luo et al. (2020) presented a comprehensive survey on steel strip surface defect inspection methods by reviewing about 120 publications. However, current surveys on surface defect inspection based on deep learning are all reviewed from the perspective of inspection accuracy, and there is a lack of surveys from the perspective of real-time.

Real-time surface defect inspection can detect defective products in time, which is of great significance for improving production efficiency. Therefore, from the perspective of real-time inspection, this paper reviews current deep learning-based surface defect inspection methods according to different industrial scenarios and different types of surfaces. In addition, this paper introduces commonly used metrics for evaluating surface defect inspection models and publicly available defect datasets, and discusses the challenges faced by deep learning-based surface defect inspection methods.

The rest of this paper is organized as follows: Sect. 2 reviews deep learning-based real-time surface defect inspection methods. The real-time defect inspection methods based on image classification, object detection and semantic/instance segmentation in different industrial scenarios are presented. Section 3 presents common metrics for evaluating surface defect inspection models, including confusion matrix, Class Activation Mapping,

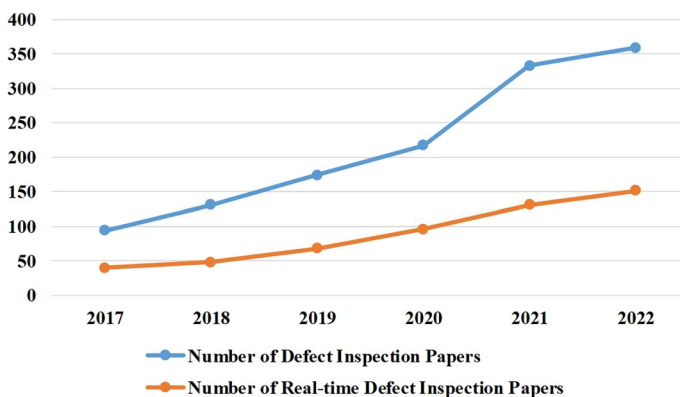


Fig. 1 In the past 5 years, the number of surface defect inspection papers and the number of real-time surface defect inspection papers in several industry-related journals

accuracy, mean Average Precision, etc. Section 4 presents publicly available surface defect datasets in various industrial scenarios. Section 5 discusses the challenges faced by deep learning-based surface defect inspection methods, including the acquisition of surface defect datasets, balancing the accuracy and speed of inspection models, and the application in industrial environments. Section 5 concludes this paper, and the future development trend of surface defect inspection is prospected.

2 Real-time surface defect inspection methods based on deep learning

Deep learning-based methods have powerful feature extraction capabilities, and can automatically extract features by learning a certain number of labeled samples. With the rapid development of deep learning and the improvement of device computing power, more and more deep learning methods have been applied to industry, and deep learning-based methods have become the mainstream method for surface defect inspection. Currently, surface defect inspection methods based on deep learning are mainly divided into three categories according to different task objectives: defect classification, defect detection and defect segmentation. As shown in Fig. 2, defect classification methods can detect whether there is a defect and the type of the defect in the obtained product image. For defect detection methods, on the basis of defect classification methods, the position of the defect can be located in the form of a rectangular frame. For defect segmentation methods, in addition to

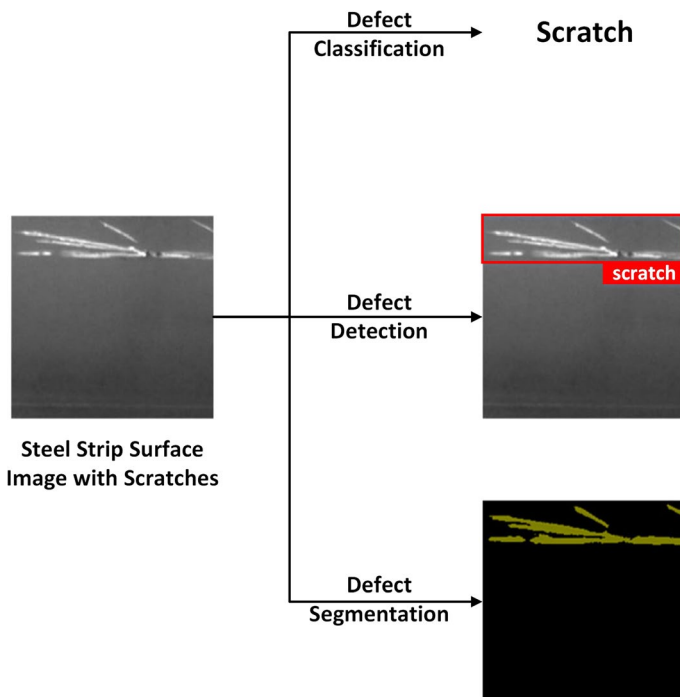


Fig. 2 Three different types of deep learning-based surface defect inspection methods

the type of defect, the defect area can be segmented from the background at the pixel level. Defect classification, defect detection, and defect segmentation correspond to image classification, object detection, and semantic/instance segmentation tasks in computer vision, respectively.

In non-real-time surface defect classification tasks, the most important is the classification accuracy, while the classification speed is ignored. With the emergence of networks such as AlexNet (Krizhevsky et al. 2012), visual geometry group (VGG) (Simonyan and Zisserman 2014), GoogleNet (Szegedy et al. 2015), ResNet (He et al. 2016), DenseNet (Huang et al. 2017), etc., the accuracy of defect classification networks has been continuously improved. However, these networks are not suitable for real-time defect classification tasks due to the slow inspection speed. In real-time surface defect classification, in addition to accuracy, inspection speed is also an important factor to consider. The appearance of lightweight classification networks, such as SqueezeNet (Iandola et al. 2016), MobileNet (Howard et al. 2017), MnasNet (Tan et al. 2019), EfficientNet (Tan and Le 2019), etc., enables real-time surface defect classification. After these lightweight networks were proposed, they have been applied to various classification tasks (Huang et al. 2020; Wentao et al. 2020; Zhu et al. 2021), proving their generality. These lightweight classification networks have also been widely used in real-time surface defect classification tasks in industry, and achieved satisfactory classification results. The defect classification method is the simplest method in deep learning-based surface defect inspection methods. Defect classification methods only need to detect whether there is a defect in the image and the type of the defect. The inspection process is shown in Fig. 3. Defect classification methods are methods that obtain the least defect information. For those scenarios that require more surface defect information (He et al. 2019a; Liu et al. 2020a; Shen et al. 2021), defect classification methods are difficult to meet the application requirements.

At present, mainstream surface defect detection algorithms are mainly divided into one-stage algorithms and two-stage algorithms. You only look once (YOLO) series (Redmon et al. 2016; Redmon and Farhadi 2017, 2018; Bochkovskiy et al. 2020) and single shot multibox detector (SSD) series (Liu et al. 2016; Fu et al. 2017; Li and Zhou 2017; Shen et al. 2017) are representatives of one-stage algorithms, while Region-CNN (R-CNN) series (Girshick et al. 2014; Girshick 2015; Ren et al. 2015) are representatives of two-stage algorithms. One-stage defect detection algorithms have faster detection speed, while two-stage defect detection algorithms have higher detection precision. Therefore, one-stage defect detection algorithms are more suitable for real-time surface defect detection. When the two-stage defect detection algorithm is applied to real-time surface defect detection, its backbone network is often replaced with a lightweight network to improve the inspection speed. Compared with defect classification methods, defect detection methods can not only

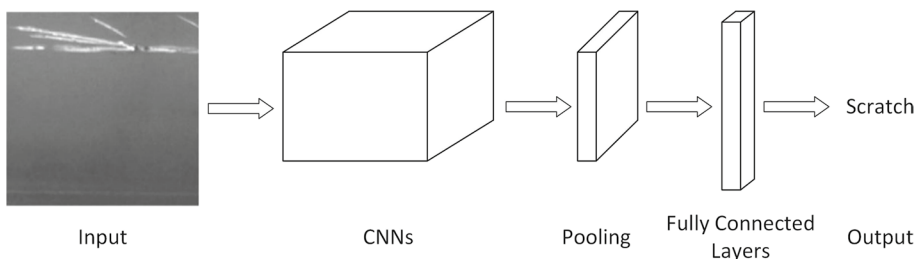
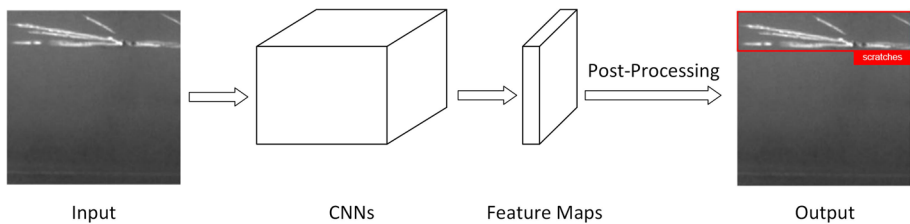


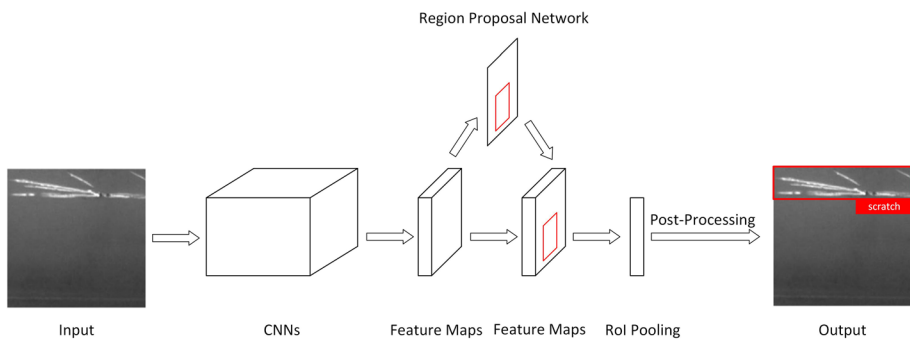
Fig. 3 Process of surface defect classification methods based on image classification

classify the defect, but also locate the defect in the image, as shown in Fig. 4. In addition, defect detection methods can detect multiple defects in one image. While defect detection methods can obtain more defect information, the network structure becomes more complex. During model training, defect detection methods have higher requirements on labels of training samples, requiring not only defect category information, but also defect location information.

Current representative defect segmentation networks are fully convolutional network (FCN) (Long et al. 2015), SegNet (Badrinarayanan et al. 2017), U-Net (Ronneberger et al. 2015), DeepLab (Chen et al. 2014, 2017a, 2017b), Mask RCNN (He et al. 2017), etc. Due to the segmentation speed, FCN and U-Net are more suitable for real-time defect segmentation, while SegNet, DeepLab and Mask RCNN are still not applied to real-time defect segmentation. Compared with defect detection methods, defect segmentation methods can detect surface defects at the pixel level, and obtain more detailed defect information, such as the shape and area of the defect, as shown in Fig. 5. Defect segmentation can obtain not only the category and location of defects, but also the severity of defects, which can be used to achieve quality classification of industrial products. However, defect segmentation methods are limited by datasets in practical applications. Compared with defect classification methods and defect detection methods, defect segmentation methods require pixel-level label information, which is expensive to label. The high cost of sample labeling is one of the key factors hindering the application of defect segmentation methods to industrial scenarios. Table 1 compares different deep learning-based surface defect inspection methods.



(a) One-stage surface defect detection



(b) Two-stage surface defect detection

Fig. 4 Process of surface defect detection methods based on object detection

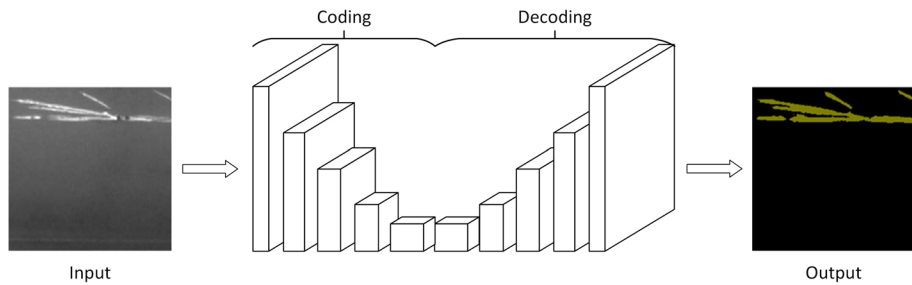


Fig. 5 Process of surface defect segmentation methods based on semantic/instance segmentation

In industry, surface defects of industrial products in different scenarios have different characteristics. Relevant researchers have proposed different surface defect inspection methods according to different defect characteristics. From the perspective of real-time inspection, this paper reviews the current surface defect inspection methods for different types of surfaces in industry, as shown in Fig. 6, including metal strip surface, metal product surface, textured surface, other types of surfaces, and general surface. Next, the defect inspection methods for each type of surface are reviewed in detail.

2.1 Real-time metal strip surface defect inspection methods

Real-time surface defect inspection methods based on deep learning have been widely used in metal strip production, mainly including steel strip and aluminum strip. Surface defects of metal strip have intraclass difference and interclass similarity, that is, defects of the same type may be greatly different, while defects of different types may have little difference. Additionally, image quality may be affected by illumination during image acquisition, and the production speed of metal strip is fast, resulting in motion blur and camera noise in the acquired images. Therefore, when designing a surface defect inspection method for metal strip, it is necessary to fully consider the characteristics of metal strip surface defects and the impact of the production environment during image acquisition. Next, this paper reviews the surface defect inspection methods of metal strip based on defect classification, defect detection and defect segmentation, respectively.

2.1.1 Defect classification

Fu et al. (2019) used the pre-trained SqueezeNet as the backbone architecture of the steel strip surface defect classification network and optimized the network. First, in fine-tuning, a higher learning rate is set for shallower layers to better extract defect features. Then, new network branches are added to extract the features of multiple receptive fields. Then, the features of multiple receptive fields are fused to improve the scale invariance of high-level features and the capability to detect defects of various sizes. This method achieves 100% classification accuracy with an inspection speed of 8.0ms per image on the NEU-CLS dataset (Song and Yan 2013). They also constructed an augmented test set doped with various noises to evaluate the robustness of their method. To detect surface defects of steel strip and reduce the dependence on the number of training samples, Liu et al. (2020b) designed a CNN model based on multi-scale images. Images of different scales are separately input into multiple CNNs with the same structure for parallel computation, and the

Table 1 Comparison of deep learning-based surface defect inspection methods

Method	Representative networks	Label information	Advantages	Disadvantages
Defect classification	VGG, SqueezeNet, MobileNet	Classification	The network structure is simple and easy to implement. The inspection speed is fast, and the dataset labeling cost is low.	The obtained defect information is little, and it is difficult to meet the requirements of product quality control in some industrial scenarios.
Defect detection	YOLO, SSD, R-CNN	Classification, location	The category, location and size information of defects can be obtained, which basically meets the requirements of product quality control, and the cost of dataset labeling is moderate.	The network structure is complex, which requires a trade-off between the detection precision and speed.
Defect segmentation	FCN, U-Net, Mask RCNN	Segmentation	Detailed defect information can be obtained, which can provide reliable basis for product quality control.	Dataset labeling cost is high, and the defect quantity information cannot be obtained.

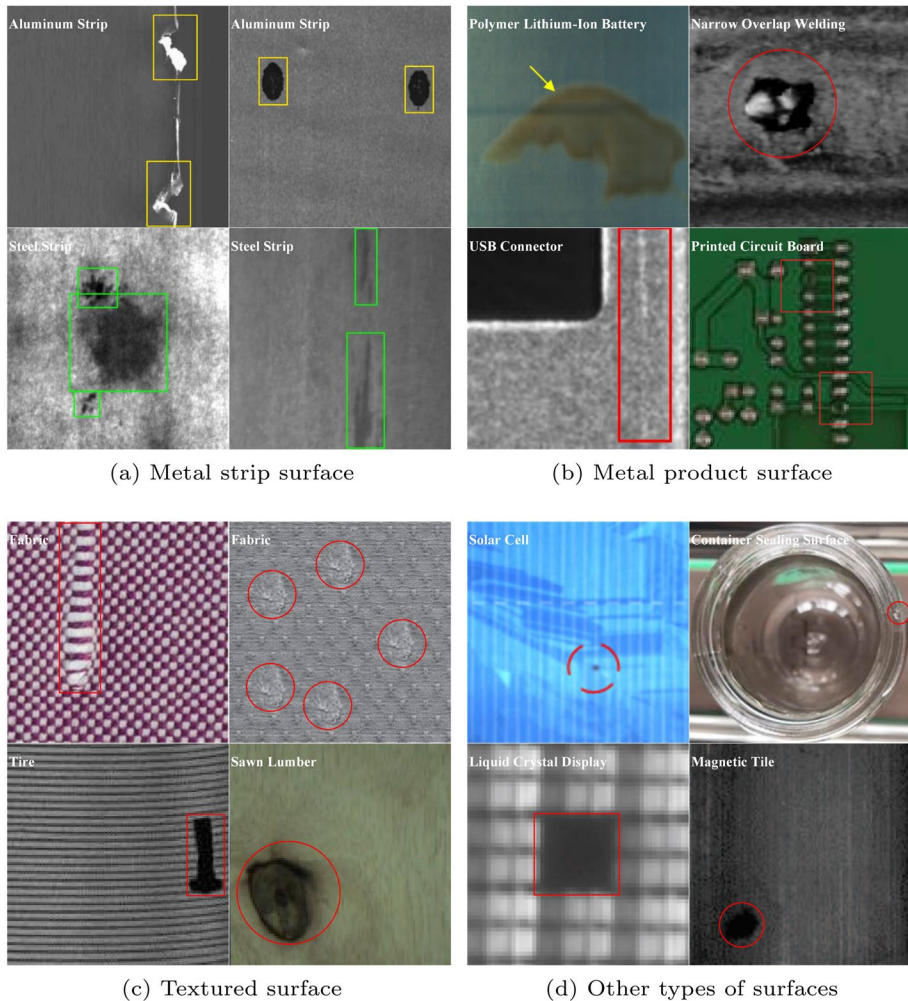


Fig. 6 Different types of surfaces and their defects in industry

output of each CNN is concatenated to predict the type of defect. Compared with other high-performance networks (MobileNet, ThumbNet (Zhao and Ghanem 2020), VGG-16, and PyramidNet (Han et al. 2017)), this CNN model achieves the highest classification accuracy of 98.89% and an inspection speed of 5.58 ms per image while using only 20% of the training samples in the NEU-CLS dataset.

2.1.2 Defect detection

Defect detection method is the most commonly used method for surface defect inspection of metal strip, mainly including one-stage defect detection method and two-stage defect detection method. For one-stage defect detection methods, Li et al. (2018a) constructed a fully convolutional YOLO with 27 convolutional layers to detect steel strip surface defects. The first 25 convolutional layers of this YOLO are used to extract surface defect features

of steel strip, and the last two convolutional layers are used to predict defect categories and bounding boxes. To improve inspection precision, multi-scale images are used to train the model, and feature pyramid network (FPN) (Lin et al. 2017) is introduced. This YOLO model achieves 97.55% mean average precision (mAP) and 95.86% recall on six types of defects, and achieves 99% detection rate with a speed of 83 frames per second (FPS). On the basis of YOLOv4, Ma et al. (2022b) proposed a more lightweight method for detecting surface defects of aluminum strip. Based on MobileNet, they redesigned the backbone network of YOLOv4, and added channel attention and spatial attention mechanisms. In addition, they optimized the Path Aggregation Network (PANet) (Liu et al. 2018) in YOLOv4 based on spatial pyramid pooling (SPP) (He et al. 2015b) and depthwise separable convolution. The improved YOLOv4 reduces the training parameters from 64,429,405 to 11,729,069, reduces the model size by 64%, and achieves 96.28% mAP and 21.537 FPS inspection speed. Ma et al. (2022a) also overcame YOLOv4's shortcomings of low computational efficiency and high computational cost. Based on the depthwise separable convolution and the lightweight convolution block in GhostNet (Han et al. 2020), they compressed the model scale and reduced computational cost. Besides, channel and spatial attention mechanisms are introduced to enhance YOLOv4's feature extraction capability. Furthermore, based on passthrough and pixelshuffle, they proposed a new sampling method aimed at better feature mapping. Compared with the original YOLOv4, the model scale of the improved YOLOv4 is reduced by 80%, and the inspection speed is increased by 3 times. The improved YOLOv4 achieves 94.68% mAP and 20.749 FPS inspection speed on the aluminum strip surface defect inspection task.

For two-stage defect detection methods, He et al. (2019) proposed a multi-level feature fusion network with high computational efficiency and strong generalization capability, and then used region proposal network (RPN) to slide on the multi-level feature maps to generate candidate boxes for defect classification and bounding box regression. The pre-trained ResNet-34 and ResNet-50 are respectively used as the backbone network to meet the different requirements of inspection precision and speed. This network is able to achieve an inspection speed of 20 ft/s while keeping the mAP at 70%. Ren et al. (2018) implemented a Slighter Faster R-CNN based on Faster R-CNN and depthwise separable convolution. Compared with Faster R-CNN, the inspection speed of Slighter Faster R-CNN is increased by 3 to 4 times. Slighter faster R-CNN adds center loss (Wen et al. 2016) to the original loss function, which improves the capability to distinguish different categories of defects. Slighter Faster R-CNN achieves 98.32% mAP and 20 FPS inspection speed on the acquired steel strip surface defect dataset.

In recent years, object detection algorithms based on Anchor-Free have also been applied to the real-time metal strip surface defect inspection task. Wang et al. (2022b) proposed a real-time detection network that balanced speed and precision to detect surface defects for steel strip. Based on ResNet-dcn and ResNet-18, they designed the backbone network ResNet18-dcn. To improve the inspection performance of surface defects with large intraclass difference and high interclass similarity, a cross-layer connection module is introduced into the backbone network, and a pyramid feature fusion module that fuses multiple high-level features is proposed. To achieve higher inspection speed, they adopted the Anchor-Free detection method to generate bounding boxes based on the center points. The proposed network achieves 80.0% mAP and an inspection speed of 64 FPS on the NEU-DET dataset. Yu et al. (2021) proposed a fully convolutional one-stage detection network based on channel attention and bidirectional feature fusion for steel strip surface defect inspection. To solve the problem of information loss in the feature extraction process, they designed a channel attention module, which allowed the network to utilize global features

to adaptively emphasize features with missing information and suppress other features. They proposed a bidirectional feature fusion network to replace FPN, making each layer retain semantic and spatial information. This detection network achieves 76.68% mAP and 18 FPS on the NEU-DET dataset.

2.1.3 Defect segmentation

At present, defect segmentation methods are rarely used in metal strip surface defect inspection. To solve the problem of blurred edges during defect segmentation, Song et al. (2020a) proposed a segmentation network based on encoder-decoder structure and residual network to detect surface defects of steel strip. The segmentation network consists of three parts: encoder network, decoder network and residual refinement network. In the encoder network, a fully convolutional network is used to extract multi-level features, and a channel weighted block is designed to make the network pay more attention to defective regions. In the decoder network, a residual decoder block is proposed and combined with the channel weighted block to recover the encoded image information. The residual refinement network adopts an encoder-decoder network structure to optimize the roughly segmented defect edges. On the SD-saliency-900 dataset (Song et al. 2020b), the proposed segmentation network can accurately segment the edges of defects, and achieve 84.16% mean Intersection over Union (mIoU) and 27 FPS inspection speed.

Among the real-time surface defect inspection methods for metal strip, the defect detection method is the most commonly used method. For each type of surface defect inspection method, high accuracy and fast inspection speed cannot be achieved at the same time, and higher accuracy often leads to a decrease in inspection speed. Table 2 summarizes the performance and inspection speed of current real-time surface defect inspection methods for metal strip.

2.2 Real-time metal product surface defect inspection methods

Defect inspection on the metal product surface is an important part of surface defect inspection in industry, including laser powder bed fusion, welding, printed circuit board, etc. Metal product surface defects not only have the characteristics of metal strip surface defects, but also are more complex than metal strip surface defects. The boundary between metal product surface defects and the background is not obvious, and the defect scale is varied. In addition, there are interference factors such as stains, shadows, uneven illumination, etc. in the acquired metal product images. Compared with defect inspection methods on the metal strip surface, defect inspection methods on the metal product surface usually use more complex network structures. Current real-time surface defect inspection methods for metal products are mainly divided into defect classification methods and defect detection methods, and lack of defect segmentation methods. Next, each type of method is reviewed.

2.2.1 Defect classification

To detect the welding defects of power battery, Yang et al. (2020c) used the pre-trained SqueezeNet as the classification network, and modified the structure of the network. They reduced the number of convolution kernels to prevent overfitting while reducing the model parameters, and moved the max pooling layer to the back of Fire 3 and Fire 5, which

Table 2 Performance and inspection speed of current real-time surface defect inspection methods for metal strip

Method	Reference	Industrial scenario	Number of training samples	Image size	Performance	Speed (FPS)
Defect classification	Fu et al. (2019)	Steel strip	80% of NEU-CLS dataset	200 × 200	Accuracy: 100%	125
	Liu et al. (2020b)	Steel strip	20% of NEU-CLS dataset	200 × 200	Accuracy: 98.89%	179.2
	Li et al. (2018a)	Steel strip	3455	300 × 300	mAP: 97.55%	83
	Ma et al. (2022b)	Aluminum strip	5530 for training and validation	608 × 608	mAP: 96.28%	21.537
	Ma et al. (2022a)	Aluminum strip	70% of 6820 samples	608 × 608	mAP: 94.68%	20.749
	He et al. (2019c)	Steel strip	70% of NEU-DET dataset	200 × 200	mAP: 70.0%	20 f/s
	Ren et al. (2018)	Steel Strip	3455	600 × 600	mAP: 98.32%	20
Defect detection	Wang et al. (2022b)	Steel strip	70% of NEU-DET dataset	200 × 200	mAP: 80.0%	64
Defect segmentation	Yu et al. (2021)	Steel strip	80% of NEU-DET dataset	200 × 200	mAP: 76.68%	18
	Song et al. (2020a)	Steel strip	540 source images and 270 image disturbed by noise	200 × 200	mIoU: 84.16%	27

reduced the computation by about 2.4 times compared to the original SqueezeNet. The improved SqueezeNet achieves 99.57% and 95.58% classification accuracy at 4.9 ms per image on the collected 2-classification and 7-classification datasets, respectively. To detect two types of surface defects, delamination and splatter, generated during the laser powder bed fusion process, Baumgartl et al. (2020) designed a network structure with only three layers of convolution. They introduced depthwise separable convolution to reduce parameters, and used dropout to prevent overfitting. This designed network achieves 96.80% accuracy while satisfying real-time inspection. Scime and Beuth (2018) designed a multi-scale CNN based on pre-trained AlexNet to detect surface defects in the laser powder bed fusion process. The inputs to the CNN are image patches of three different scales. Among them, the first-level image patch is a square block manually selected by human experts from the image, the second-level image patch is an image block with the same center as the first-level image patch but with a larger resolution, and the third-level image patch is the entire image after interpolation. If there is an anomaly in the first-level image patch, the image will be marked as a defect image. The multi-scale CNN achieves real-time inspection and 97.0% accuracy. Ma et al. (2019) proposed an optimization-based CNN to detect surface defects for polymer Lithium-Ion battery. Since different feature maps have different importance in the surface defect classification task, trainable weight parameters are introduced into the CNN to indicate the importance of different feature maps, which are used to quickly discover and effectively extract important features for defect classification. In addition, flower pollination algorithm is used to find the optimal learning rate, which improves the stability and speed of network training. The optimized CNN achieved 98.60% accuracy with an inspection time of no more than 0.3 s per image. Deng et al. (2018) obtained the defect image by aligning the printed circuit board defect image with the template image and cropping, and proposed a lightweight CNN. The lightweight CNN achieves real-time defect classification with 99.70% accuracy on the defect dataset.

2.2.2 Defect detection

For defect detection methods, Yang et al. (2020a) decomposed the defect detection task into multiple defect classification tasks. They used background segmentation and template matching techniques to separate the universal serial bus (USB) connector from the background and divided it into multiple patches, and then input each patch into the pre-trained SqueezeNet to predict the defect category. They added a multi-receptive field fusion structure to SqueezeNet to improve the detection ability of defects with different scales. This method achieves 95.3% accuracy, and the inspection speed exceeds 100 FPS. Miao et al. (2022) proposed two two-stage models for the inspection of narrow overlap welding defects based on eddy current signals and laser images, respectively. For the two-stage eddy current detection model, in the first stage, a CNN with large sliding windows is used to detect defects in larger regions, and in the second stage, for those regions containing defects, a CNN with small sliding windows is used to detect the location and class of defects. For the two-stage laser detection model, in the first stage, the image is segmented and each image patch is detected to determine whether there is a defect, and in the second stage, the defect in the image patch is classified. The eddy current detection model and the laser detection model achieve 96.94% and 96.92% accuracy respectively, and the time of one inspection is 2.4 s and 3.2 s respectively, which meet the real-time requirements. Based on structural similarity (SSIM) (Wang et al. 2004) and MobileNet-V3 (Howard et al. 2019; Xia et al. 2019) proposed SSIM-NET for printed circuit board surface inspection. In the first stage,

SSIM is used to detect regions that may have defects, and SSIM is not easily affected by illumination, camera noise, etc. In the second stage, MobileNet-V3 classifies potentially defective regions and adds Focal Loss as an auxiliary loss function to accelerate model convergence. SSIM-NET achieves 97.06% mAP and the inspection speed reaches 60 FPS. Lan et al. (2021) proposed an improved YOLOv3-based surface defect detection method for printed circuit boards. To improve the inference speed of YOLOv3, they combined Batch Normalization to the convolutional layer. Aiming at the problem that the objective function and evaluation metric of YOLOv3 are not unified, they used Generalized Intersection over Union (GIoU) (Rezatofighi et al. 2019) to improve the detection ability for small defects on printed circuit board surface. The improved YOLOv3 achieves 92.13% mAP and an inspection speed of 63 FPS.

For real-time surface defect inspection methods for metal products, there is still a lack of defect segmentation methods. Table 3 summarizes the performance and inspection speed of current real-time defect classification methods and defect detection methods for metal product surface.

2.3 Real-time textured surface defect inspection methods

Textured surface is one of the most common surfaces in industry, such as fabric, sawn lumber, tire, etc. Textured surface generally has the characteristics of complex and diverse textures. The textures of defect and defect-free areas have similar colors and styles, and there is no obvious boundary between defect areas and defect-free areas. Even the texture of the same industrial product may vary greatly, for example, the texture between the sidewall and tread of a tire is different, which requires defect inspection methods to be robust to texture differences. In addition, the incomplete texture defects caused by cutting also add difficulties to the defect inspection of textured surface. This section provides an overview of current research on real-time defect inspection of textured surface.

2.3.1 Defect classification and detection

At present, there are few real-time defect inspection methods for textured surface based on defect classification and defect detection. For the defect classification method, based on the existing SurfNet (Arikan et al. 2019; Tsai and Chen 2020) designed SurfNetV2 with faster inspection speed and higher inspection accuracy for real-time calcium silicate board surface defect inspection. They reduced the size of the convolution kernel in SurfNet, and replaced the PReLU activation function (He et al. 2015a) in the residual block with ReLU to improve the inspection speed. And they increased the number of SurfNetV2 blocks in the network to improve the feature extraction capability, solving the problem of SurfNet's poor performance on certain textured surface defects. SurfNetV2 achieves 99.90% and 99.75% accuracy on the calcium silicate board surface defect dataset and NEU-CLS dataset, respectively, and achieves an inspection speed of 199.38 FPS on images with the resolution of 128×128 pixels. Dlamini et al. (2022) used YOLOv4 to locate fabric defects, achieving 95.3% mAP and an inspection speed of 34 FPS. To improve the localization accuracy of sawn lumber surface defects, Tu et al. (2020) introduced a Gaussian function into YOLOv3 to predict the coordinates of the bounding box. The improved YOLOv3 achieves real-time inspection, while increasing True Positive and True Negative and reducing False Positive and False Negative. The improved YOLOv3 achieves 78.04%

Table 3 Performance and inspection speed of current real-time surface defect inspection methods for metal product

Method	Reference	Industrial scenario	Number of training samples	Image size	Performance	Speed (FPS)
	Yang et al. (2020c)	Welding of power battery	2-Classification: 17,981 7-Classification: 9912	224 × 224	2-Classification Accuracy: 99.57% 7-Classification Accuracy: 95.58%	204.1
Defect classification	Baumgartl et al. (2020)	Laser powder bed fusion	4313	224 × 224	Accuracy: 96.80%	Real-time
	Scime and Beuth (2018)	Laser powder bed fusion	–	900 × 900	Accuracy: 97.0%	Real-time
	Ma et al. (2019)	Polymer lithium-ion battery	18,860	219 × 219	Accuracy: 98.60%	> 3.3
	Deng et al. (2018)	Printed circuit board	–	–	Accuracy: 99.70%	Real-time
	Yang et al. (2020a)	USB connector	6000	200 × 200	Accuracy: 95.3%	> 100
	Miao et al. (2022)	Narrow overlap welding	2991	–	Accuracy (Eddy Current): 96.94% Accuracy (Laser): 96.92%	Eddy current: 0.4 Laser: 0.3
Defect detection	Xia et al. (2019)	Printed circuit board	–	–	mAP: 97.06%	60
	Lan et al. (2021)	Printed circuit board	–	–	mAP: 92.13%	63

and 88.85% mAP, and inspection speeds of 88.07 FPS and 94.49 FPS on the rubber lumber and pine lumber datasets, respectively.

2.3.2 Defect segmentation

The defect segmentation method is the most commonly used method for real-time defect inspection on textured surface. Based on U-Net and MobileNetV2, Jing et al. (2022) proposed a lightweight and efficient Mobile-Unet for end-to-end fabric surface defect segmentation, in which the pre-trained MobileNetV2 was used as the encoder. For fabric surface defect inspection, defect-free samples are easy to collect, while defect samples are difficult to collect, resulting in an imbalance in the number of samples, and the small area of fabric defects results in an imbalance in the number of defect pixels and background pixels. To address the problem of sample imbalance, they used an improved cross-entropy loss based on median frequency balancing loss function (Kampffmeyer et al. 2016). Mobile-Unet achieves 70.0% and 92.0% mIoU on the Fabric Images dataset (Tsang et al. 2016) and Yarn-dyed Fabric Images dataset, respectively, with an inspection speed of 7.05ms per image. To solve the problem of lacking samples, Huang et al. (2021b) proposed a fabric defect segmentation method consisting of a segmentation network and a decision network. In the first stage, the segmentation network locates the defects in the image, and considers each pixel as a training sample to increase the number of samples. In the second stage, samples are classified into two categories (defect samples and defect-free samples). During network training, the two networks are trained separately. The proposed method only requires nearly 50 defect samples to complete the network training and achieves 25 FPS inspection speed. Zheng et al. (2020) proposed a concise semantic segmentation network consisting of a semantic segmentation module and a defect classification module for tire surface defect inspection. The semantic segmentation module is implemented based on an auto-encoder, which is used to perform binary classification on each pixel of the input image. Then, the defect area is cropped and used as the input of the defect classification module to predict the type of defect. The concise segmentation network achieves 77.34% mIoU and an inspection speed of 0.132 s per image on the collected dataset.

For real-time textured surface defect inspection methods, the most commonly used methods are defect segmentation methods, and there are fewer defect classification and defect detection methods. Table 4 summarizes the performance and inspection speed of current real-time defect inspection methods for textured surface.

2.4 Real-time defect inspection methods for other types of surfaces

In addition to a review of real-time defect inspection methods for the most common surfaces (metal strip surface, metal product surface, and textured surface) in industry, this paper also reviews the research on other types of surfaces.

2.4.1 Defect classification

Wang et al. (2019a) designed a CNN based on the inverted residual block (Sandler et al. 2018) to detect surface defects on wine bottles. The inverted residual block effectively reduces the network scale and reduces the computational burden without affecting the accuracy. This CNN achieves 99.60% accuracy at 47.60ms per image. Chen et al. (2020) analyzed the defect features of solar cells and found that different defects have different

Table 4 Performance and inspection speed of current real-time defect inspection methods for textured surface

Method	Reference	Industrial scenario	Number of training samples	Image size	Performance	Speed (FPS)
Defect classification	Tsai and Chen (2020)	Calcium silicate board	–	128 × 128	Accuracy: 99.90%	199.38
	Dlamini et al. (2022)	Fabric	–	650 × 650	mAP: 95.3%	34
Defect detection	Tu et al. (2020)	Sawn lumber	Rubber Lumber: 60% of 5840 samples Pine Lumber: 60% of 1108 samples	320 × 320	mAP (Rubber Lumber): 78.04% mAP (Pine Lumber): 88.85%	Rubber Lumber: 88.07 Pine Lumber: 94.49
	Jing et al. (2022)	Fabric	Fabric Images: 600 Yarn-dyed Fabric Images: 890	256 × 256	mIoU (Fabric Images): 70.0% mIoU (Yarn-dyed Fabric Images): 92.0%	141.8
	Huang et al. (2021b)	Fabric	50	256 × 256	Only nearly 50 defect samples are required to complete network training.	25
Defect segmentation	Zheng et al. (2020)	Tire	–	–	mIoU: 77.34%	7.6

contrasts in the same spectrum band, and the same defect has different contrasts in different spectrum bands. Therefore, they designed a multi-spectral CNN. The original image is divided into more and smaller image patches by sliding windows as the input of the network, and then the image patches of different spectra are input into different CNNs, respectively. The output of each CNN is concatenated for defect classification. The multi-spectral CNN can complete the defect inspection within 50ms, and the accuracy reaches 94.30%.

2.4.2 Defect detection

Zhang et al. (2020b) combined MobileNet with SSD and proposed a real-time high-accuracy vehicle body paint surface defect detection model, MobileNet-SSD. Based on the characteristics of paint defects, they improved the feature layer of MobileNet-SSD and optimized the bounding box matching strategy. MobileNet-SSD achieves over 95% accuracy while satisfying real-time inspection, which is 10% faster than traditional SSD. Li et al. (2018b) proposed a container sealing surface defect inspection algorithm based on MobileNet and SSD, which achieved over 95% detection rate with an inspection speed of 0.12 s per image. Çelik et al. (2020) used three different detectors (RetinaNet, YOLOv3 and M2Det) for liquid crystal display (LCD) surface defect inspection. Among them, the RetinaNet-based detector includes a backbone network and two task-specific networks, which are used for feature extraction, defect classification and bounding box regression, respectively. The YOLOv3-based detector adopts FPN for multi-scale detection, and the M2Det-based detector utilizes a multi-level FPN to extract multi-scale features more efficiently. The RetinaNet-based detector balances performance and speed, achieving 99.7% recall and detecting a 55-inch panel in 16 s. Based on YOLOv3-tiny, Yao and Li (2022) proposed AYOLOv3-tiny to detect surface defects of portable android device (PAD) light guide plate. Based on the overlapping pooling and spatial attention mechanism, they proposed the overlapping pooling spatial attention module to replace the traditional convolution in the YOLOv3-tiny backbone network. Overlapping pooling can use adjacent pixel information to improve accuracy, and the spatial attention mechanism can improve the feature extraction capability of the backbone network. To improve the inspection ability of large defects, the dilated convolution is integrated into the residual network, which expands the receptive field of the convolution kernel. AYOLOv3-tiny achieves 99.50% mAP and an inspection speed of 144 FPS.

2.4.3 Defect segmentation

Huang et al. (2018) proposed a saliency inspection model named MCuePush U-Net for magnetic tile surface defect segmentation. By analyzing the characteristics of blowhole defects and crack defects on grayscale images, they found five important cues for constructing saliency images. To improve the inspection speed, the region of interest cropped on the saliency image and the original image is used as the input of the saliency inspection model. The saliency model is composed of U-Net and Push network. Push network is an independent structure, and retrieves information including the locations and sizes of surface defects according to the semantic features obtained from the most down-sampled layers. MCuePush U-Net achieves 98.5% area under curve (AUC), and the inspection speed reaches 0.07 s per image. Table 5 summarizes the performance and inspection speed of current real-time defect inspection methods for other types of surfaces.

Table 5 Performance and inspection speed of current real-time defect inspection methods for other types of surfaces

Method	Reference	Industrial scenario	Number of training samples	Image size	Performance	Speed (FPS)
Defect classification	Wang et al. (2019a)	Wine bottle	70% of 332 samples	1286 × 962	Accuracy: 99.60%	21
	Chen et al. (2020)	Solar cell	—	256 × 256	Accuracy: 94.30%	> 20
	Zhang et al. (2020b)	Vehicle body paint	—	300 × 300	Accuracy: >95%	Real-time
	Li et al. (2018b)	Container sealing surface	—	300 × 300	Detection Rate: 95%	8.3
Defect detection	Çelik et al. (2020)	LCD	1492	1000 × 1000	Recall: 99.7%	55-inch panel in 16 s
Defect segmentation	Yao and Li (2022)	PAD light guide plate	—	416 × 416	mAP: 99.50%	144
	Huang et al. (2018)	Magnetic tile	—	98 × 98	AUC: 98.5%	14.3

2.5 General real-time surface defect inspection methods

In addition to designing real-time defect inspection methods for specific types of surface defects, some researchers have implemented general real-time surface defect inspection methods based on common characteristics of surface defects. In industrial environments, dust and illumination lead to low contrast between defects and background in the acquired images. Surface defects generally have the characteristics of interclass similarity and intra-class difference, and the shape and size of defects are random. General real-time surface inspection methods can achieve good inspection performance on different types of surfaces, and even achieve better performance than specific types of surface inspection methods on specific types of surfaces. For general real-time surface defect inspection methods, not only the common characteristics of surface defects, but also the interference of different surface characteristics to the inspection method needs to be considered. Next, general real-time surface defect inspection methods are reviewed.

2.5.1 Defect classification

Wang et al. (2018) carefully designed a joint inspection CNN. The joint inspection CNN contains two stages: first, the image sample is classified according to the background texture features (global frame classification), and then, whether the image contains defective regions is detected (subframe classification). The two stages are connected in series to form the entire network for surface defect inspection. For global frame classification, the entire image is used as input, and the sample class is predicted. For subframe classification, a sliding window method is used to divide the image into multiple blocks, which are respectively input into the network to predict whether the block contains defects. If a defect is detected in any of the blocks, the entire image is marked as a defect sample. The joint inspection CNN achieves 99.80% classification accuracy on the DAGM dataset (Wieler and Hahn 2007) and is able to detect 27 images per second. Arikan et al. (2019) designed a general surface defect inspection model, SurfNet, based on the VGG-16 network configuration and residual network. To make the model lightweight, they only used 10 layers in VGG-16, where the first nine layers were used for feature extraction and the last layer was used for classification. To address the imbalance in the number of samples in different categories, they used data augmentation, pix2pix (Isola et al. 2017) and CycleGAN (Zhu et al. 2017) to generate samples. SurfNet achieves 98.0% accuracy at 1.92ms per image. Gao et al. (2019) proposed a surface defect inspection method based on multi-layer information fusion. First, a local illumination normalization is performed on the sample to standardize the unbalanced illumination. Then, a three-level Gaussian pyramid is implemented, and images of different levels are generated and input into different VGG-16 networks. Finally, the outputs of different networks are fused to obtain inspection results. The proposed method achieves good performance on small-sample tasks and achieves an inspection speed of 13.74ms per image.

2.5.2 Defect detection

Cui et al. (2021) proposed a fast and accurate surface defect detection network. To solve the problem of large defect texture variation and small defect size, they designed feature retaining block (FRB) and skip densely connected module (SDCM). FRB simultaneously

acquires low-level and high-level features by fusing features of different resolutions. Unlike DenseNet, SDCM performs skip connections between multi-scale layers to enhance the information of small-scale defects. The proposed network achieves 88.8% mAP on the NEU-DET dataset, 99.1% mAP on the DAGM dataset, and 93.4% mAP on the magnetic tile dataset with an inspection speed of 31.0 FPS.

2.5.3 Defect segmentation

Dong et al. (2019) proposed a pyramid feature fusion and global context attention network (PGA-Net), with pre-trained VGG-16 as the backbone network for extracting multi-level features. Inspired by (Dollár et al. 2014; Wang et al. 2019b), a feature pyramid fusion module is proposed, which enables the fused feature maps of different scales to obtain multi-scale context information. To enhance the inner relation information between feature maps of different resolutions, a global context attention module is designed. By introducing deep supervision into the model, the training process is simplified and the speed of model optimization is accelerated. PGA-Net shows good performance on NEU-Seg dataset, DAGM dataset, magnetic tile defect dataset (Huang et al. 2018), road defect dataset (Yang et al. 2019a), and the inspection speed can reach 41–49 FPS. Oz et al. (2021) proposed a nested autoencoder approach for surface defect inspection. First, for a defect image, a one-dimensional vector is obtained through the encoder, denoted as a defect vector. Then, a defect-free vector is predicted through an autoencoder composed of fully connected layers, and a defect image and a defect-free image are generated according to the defect vector and the defect-free vector, respectively. Finally, an error map is obtained based on the newly generated two images for defect segmentation. The approach achieves 78% mIoU at 88ms per image on the MVTecAD dataset (Bergmann et al. 2019). Table 6 summarizes the performance and inspection speed of current general real-time surface defect inspection methods.

2.6 Summary

Current real-time surface defect inspection methods in industry are mainly divided into three types: defect classification, defect detection, and defect segmentation. From the perspective of inspection speed, Table 7 reviews the different types of surface defect inspection methods in industry.

In real-time surface defect inspection, the most commonly used method is the defect detection method, especially on the metal strip surface and metal product surface. But in real-time textured surface defect inspection, the most commonly used method is the defect segmentation method. Therefore, in actual industrial scenarios other than textured surface, when choosing a real-time defect inspection method, the defect detection method should be given priority. In defect detection, the method based on YOLOv3 or YOLOv4 is the most commonly used method, which achieves high inspection speed while achieving high-precision inspection. Lightweight networks, e.g., SqueezeNet, MobileNet, have been widely used in defect classification, defect detection and defect segmentation. The improved defect inspection model based on the lightweight network can reduce the size of the model, and increase the inspection speed to achieve real-time inspection. In addition, for the lack of training samples, Liu et al. (2020b), Huang et al. (2021b) and Gao et al. (2019) proposed real-time surface defect inspection methods based on a small number of samples. For large images (image size ≥ 1000), Wang et al. (2019a), Çelik et al. (2020) and Oz et al. (2021) proposed real-time defect inspection methods with large images as input.

Table 6 Performance and inspection speed of current general real-time surface defect inspection methods

Method	Reference	Industrial scenario	Number of training samples	Image size	Performance	Speed (FPS)
Defect classification	Wang et al. (2018)	DAGM	70% of DAGM dataset	512 × 512	Accuracy: 99.80%	27
	Arikan et al. (2019)	–	–	128 × 128	Accuracy: 98.0%	520.8
	Gao et al. (2019)	–	A small number of samples	128 × 128	Perform well on small-sample tasks.	72.8
Defect detection	Cui et al. (2021)	Steel strip	70% of NEU-DET datasets	512 × 512	mAP: 88.8%	31.0
		DAGM	1046	512 × 512	mAP: 99.1%	
		Magnetic tile	70% of 1344 samples	512 × 512	mAP: 93.4%	41–49
		Steel strip	3630	192 × 192	mIoU: 82.15%	
		DAGM	3550	192 × 192	mIoU: 74.78%	
Defect segmentation	Dong et al. (2019) Oz et al. (2021)	Magnetic tile	2840	192 × 192	mIoU: 71.31%	11.4
		Road defect	6000	192 × 192	mIoU: 79.54%	
		MVTec AD	3629	1024 × 1024	mIoU: 78%	

Table 7 Real-time performance of current deep learning-based real-time defect inspection methods on different types of surfaces

Surface type	Industrial scenario	Speed (FPS)		
		Defect classification	Defect detection	Defect segmentation
Metal strip surface	Steel strip	125–179.2	18–83	27
	Aluminum strip	–	20.7	–
	Welding	204.1	0.3–0.4	–
	Polymer lithium-ion battery	> 3.3	–	–
Metal product surface	USB connector	–	> 100	–
	Printed circuit board	–	60–63	–
	Calcium silicate board	199.4	–	–
	Sawn lumber	–	88.07–94.49	–
Textured surface	Fabric	–	34	25–141.8
	Tire	–	–	7.6
	Wine bottle	21	–	–
	Solar cell	> 20	–	–
	Container sealing surface	–	8.3	–
	PAD light guide plate	–	144	–
Other types of surfaces	Magnetic tile	–	–	14.3
General surface	–	27–520.8	31	11.4–49

3 Evaluation metrics

This section summarizes some metrics for evaluating the performance of real-time surface defect inspection methods.

3.1 Confusion matrix

Confusion matrix is a common metric for evaluating the performance of surface defect classification models, which can be used for binary classification and multi-classification tasks. The confusion matrix is a matrix that records the number of predicted samples and the number of true samples for each class. The values on the diagonal line represent the number of samples where the predicted value is the same as the true label, that is, the number of correctly classified samples, and the values deviated from the diagonal line are the number of misclassified samples. The confusion matrix enables visualization of model performance, providing an intuitive understanding of the behavior of classification models.

3.2 Accuracy, precision, recall and F1-measure

Accuracy is the most common evaluation metric, which can be used to evaluate the overall performance of the model. However, for defect detection models, the most common problem is that the number of defect samples in each category is unbalanced. Accuracy only is not enough to evaluate the model, and precision, recall and F1-measure are also needed as the evaluation metrics for the model. The definition of each metric is as follows:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \\ \text{F1-measure} &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

where TP (True Positive) is the number of correctly classified positive samples, TN (True Negative) is the number of correctly classified negative samples, FP (False Positive) is the number of misclassified negative samples, and FN (False Negative) is the number of misclassified positive samples. Precision measures the model's ability to not misclassify positive samples, recall measures the model's ability to find positive samples, and F1-measure can be thought of as a weighted average of precision and recall.

3.3 Class activation mapping

Class Activation Mapping (CAM) (Zhou et al. 2016) is able to visualize the focus of the CNN on each region in the image, enhancing the interpretability of the neural network. CAM is generated by weighted linear summation of feature maps output from the last convolutional and pooling layers, and up-sampling to the input image size. CAM shows the image regions that the model focuses on when classifying a certain surface defect. Surface defect inspection model can be improved by analyzing the focus of the model on each region in the image. Commonly used CAM techniques are Grad-CAM (Selvaraju et al. 2017) and Grad-CAM++ (Chattopadhyay et al. 2018).

3.4 mAP

Mean average precision (mAP) is the most commonly used evaluation metric for surface defect inspection methods based on object detection. For the defect detection task, not only defect classification but also defect localization is required. Instead of accuracy, the performance of defect detection models needs to be evaluated with mAP, which was first proposed in PASCAL visual objects classes (VOC) challenges. For a certain class of detection object, AP is obtained by calculating the underline area of the Precision-Recall curve, which is used to evaluate the performance of the detection model on a certain class of object, while mAP is the mean of multiple classes, which is used to

evaluate the overall performance of the detection model. In PASCAL VOC challenges, the definition of mAP is as follows:

$$mAP_{VOC} = \frac{\sum_{i=1}^k AP_i}{k}$$

where k is the number of classes. In COCO challenges, mAP is the mean of AP under different IoU thresholds (0.5 to 0.95 in steps of 0.05) and classes, as follows:

$$mAP_{COCO} = \frac{1}{k} \sum_{i=1}^k \frac{\sum_{j=0.5}^{0.95} AP_{ij}}{10}$$

where j is the IoU threshold every 0.05 in [0.5, 0.95].

3.5 mIoU

For defect segmentation, mIoU (mean Intersection over Union) is the most commonly used model evaluation metric. The essence of defect segmentation is to predict the defect category for each pixel in the image. Based on the intersection and union of predicted pixels and true pixels, the IoU of a certain class of defects in defect segmentation can be obtained. Predicted pixels are the sum of TP and FP, while true pixels are the sum of TP and FN. And mIoU is the mean of IoU for each class, as follows:

$$mIoU = \frac{1}{k+1} \sum_{i=0}^k \frac{TP}{FP + TP + FN}$$

where $k+1$ is the number of classes and the background.

3.6 Real-time

In industry, surface defect inspection models not only need to meet the accuracy requirements, but also need to meet the real-time requirements. For example, in the steel strip surface defect inspection, the average speed of the actual steel strip production line is about 20 m/s and the view field of a single camera is 80–100 cm, requiring the inspection model to have a speed of at least 20FPS. In addition to being able to detect all products, real-time requires that the inspection model can output inspection results in time to achieve product quality control. The real-time performance of the model can be evaluated by the time to detect each image or the number of detected images per second. In addition, the influence of various factors in the industrial environment on real-time performance need to be considered, such as hardware equipment, image acquisition equipment, etc.

For different surface defect inspection methods, not all evaluation metrics are required. Table 8 summarizes the commonly used evaluation metrics for different surface defect inspection methods.

Table 8 Evaluation metrics for different surface defect inspection methods

Evaluation metric	Method		
	Defect classification	Defect detection	Defect segmentation
Confusion matrix	✓		
Accuracy, precision, recall and F1-measure	✓	✓	✓
Class activation mapping	✓	✓	
mAP		✓	
mIoU			✓
Real-time	✓	✓	✓

4 Publicly available defect datasets

In industry, defect samples are often difficult to obtain, and the cost of sample labeling is too high. Therefore, this section summarizes some public surface defect datasets in various industrial scenarios, as shown in Table 9, where classification, detection, and segmentation in *Label* represent the label information required for the image classification task, object detection task, and semantic/instance segmentation task in computer vision, respectively. In Table 9, the paper also classifies the surface types of these surface defect datasets: metal strip (metal strip surface), metal product (metal product surface), textured (textured surface), other (other types of surfaces), and general (general surface).

NEU-CLS, Elpv-dataset, Sewer-ML, and CODEBRIM are commonly used datasets for surface defect classification. NEU-CLS contains 6 types of steel strip surface defects, each type has 300 images, and the image resolution is 200×200 . Elpv-dataset contains 2624 solar cell images, and the image resolution is 300×300 . Sewer-ML contains 18 types of sewer defects with a total of 1.3 million images. And CODEBRIM contains 5 types of concrete bridge defects with a total of 1590 high-resolution images.

NEU-DET, DeepPCB, GDXray Casting and Road Damage are commonly used datasets for surface defect detection. NEU-DET contains 6 types of steel strip surface defects, each type has 300 images, and the image resolution is 200×200 . DeepPCB contains 6 types of printed circuit board surface defects with a total of 1500 image pairs. GDXray Casting contains 2727 X-ray metal casting images with image resolution ranging from 256×256 to 768×572 . And Road Damage contains 8 types of road surface defects with a total of 9053 images, and the image resolution is 600×600 .

NEU-Seg, MT Defect, CRACK500, AITEX, RSDDs and MVTec AD are commonly used datasets for surface defect segmentation. NEU-Seg contains 3 types of steel strip surface defects, each type has 300 images, and the image resolution is 200×200 . MT Defect contains 5 types of magnetic tile surface defects with a total of 1344 images. CRACK500 contains 500 pavement crack images, and the image resolution is 2000×1500 . AITEX contains 12 types of surface defects for 7 different fabrics with 140 defect-free images and 105 defect images, and the image resolution is 4096×256 . RSDDs contains 67 express rail images and 128 common/heavy haul rail images. And MVTec AD contains 73 different types of defects with a total of 5354 images, and the image resolution is between 700×700 to 1024×1024 .

Table 9 Publicly available surface defect datasets

Dataset	Industrial scenario	Label	Surface type				
			Metal strip	Metal product	Textured	Other	General
NEU-CLS (Song and Yan 2013)	Steel strip	Classification	✓				
Elpv-Dataset (Deitsch et al. 2019)	Solar cell	Classification				✓	
Sewer-ML (Haurum and Moeslund 2021)	Sewer	Classification				✓	
CODEBRIM (Mundt et al. 2019)	Concrete bridge	Classification			✓		
NEU-DET (He et al. 2019c)	Steel strip	Detection					
DeepPCB (Tang et al. 2019)	Printed circuit board	Detection	✓				
GDXray casting (Ferguson et al. 2017)	Metal casting	Detection		✓			
Road damage (Maeda et al. 2018)	Road	Detection			✓		
NEU-seg (Dong et al. 2019)	Steel strip	Segmentation	✓				
MT defect (Huang et al. 2018)	Magnetic tile	Segmentation				✓	
CRACK500 (Yang et al. 2019a)	Pavement crack	Segmentation			✓		
AITEX (Silvestre-Blanes et al. 2019)	Fabric	Segmentation			✓		
RSDDs (Gan et al. 2017)	Rail	Segmentation		✓			
MVTec AD (Bergmann et al. 2019)	Different objects and textures	Segmentation					✓
DAGM 2007 (Wielers and Hahn 2007)	Multiple types of textures	Weak-label					✓
MVTec 3D-AD (Bergmann et al. 2021)	Different objects	Point cloud					✓

In particular, DAGM 2007 is a dataset with weak labels. DAGM 2007 contains 10 types of texture surface images, each of which has 1000 defect-free images and 150 defect images, and the image resolution is 512×512 . MVTec 3D-AD is a point cloud dataset, which contains 10 different types of objects with a total of 4147 3D images.

5 Challenges

Deep learning-based methods provide a fast and accurate end-to-end approach for surface defect inspection tasks in industry. Although deep learning methods overcome the shortcomings of traditional methods, there are still many challenges in practical applications. From the perspective of the dataset, model and application, this paper discusses the challenges faced by deep learning-based real-time surface defect inspection methods.

5.1 Dataset

Surface defect inspection methods based on deep learning require a certain number of defect samples to train the model, so that the model has the ability to automatically extract defect features. However, in the industrial production process, it is difficult to collect a certain number of defect samples. For example, the occurrence probability of steel strip surface defects is very small, and the defects are not visually obvious. For each type of defect, collecting a certain number of samples is a difficult and time-consuming process. In addition to the difficulty of collecting samples, labeling the collected samples is also a labor-intensive process. In real industrial scenarios, there may only be dozens of defect samples, and not all defect types are covered, which is far from enough for deep learning methods. Aiming at the challenge of insufficient defect samples faced by deep learning-based surface defect inspection methods in industry, four commonly used solutions are summarized below.

- (1) Data augmentation. Commonly used data augmentation methods include geometric transformation, flipping, rotation, cropping, mirroring, translation, etc. Data augmentation increases the diversity and quantity of training samples. In recent years, generative adversarial network (GAN) (Goodfellow et al. 2014) has also been used as a method of data augmentation. Le et al. (2020) used cropping and Wasserstein GAN (WGAN), respectively, to generate new decorative sheet defect samples and welding joint defect samples. He et al. (2019b) used GAN to generate steel strip surface defect samples, and proposed multi-training based on self-training and co-training to add labels to newly generated samples. Jain et al. (2020) used GAN to learn the distribution of samples, obtained defect samples with new features from a continuous sample space, and proved through experiments that on the basis of traditional data augmentation, using samples generated by GAN to train the model could further improve the performance of the defect inspection model.
- (2) Transfer learning. For deep learning models, the network parameters are large, and only using a small-scale surface defect dataset for model training will easily lead to overfitting of the model. Transfer learning is one of the most common methods to solve the problem of insufficient training samples. For transfer learning, first, the deep learning model is trained on a large-scale dataset (eg, ImageNet, COCO, etc.), so that the model has the ability to extract generalization features, and then the model is fine-tuned using

small-scale defect dataset, which enables the model to extract defect features. Fu et al. (2019) used SqueezeNet pre-trained on ImageNet as the backbone network for steel strip surface defect inspection. Due to the discrepancy between the pre-training dataset and the steel strip surface defect dataset, they fine-tune the model using a larger learning rate. Zhang et al. (2020a) proposed a new transfer learning method that divides the fine-tuning process into two steps. First, the model is fine-tuned on a dataset with high geometric similarity to the defect dataset, which is extracted from the large-scale dataset. Then, the model is fine-tuned again using the defect dataset.

- (3) **Semi-supervised and weakly supervised learning.** Semi-supervised learning only requires a small number of labeled samples and a certain number of unlabeled samples to train the model, which reduces the difficulty of labeling samples in industry. Wang et al. (2022a) proposed a semi-supervised learning defect classification method for the inspection of railway track surface defects. They used semi-supervised methods (Unsupervised Data Augmentation (Xie et al. 2020), Tri-training (Blum and Mitchell 1998), and Fix-match (Sohn et al. 2020)) as hyperparameters of the inspection model, and adopted the greedy algorithm to select the optimal semi-supervised learning method. The proposed defect inspection method can achieve 97.91% accuracy while detecting in real time. Weakly supervised methods only train the model through the classification labels of the images to achieve localization/segmentation of defects. Mehta et al. (2018) trained the model only based on the classification labels of images, achieving soiling segmentation and classification on the surface of solar panels. In the segmentation stage, they proposed Mask Fully Convolutional Neural Network for aggregating feature maps of different scales to generate candidate masks. Based on the average of the masks, the pixels in the image are divided into panel pixels and soiling pixels. In the classification stage, they extracted soiling areas based on the results of the segmentation stage, and then proposed Weakly supervised Neural Network to classify soiling types. The proposed weakly supervised learning-based method achieves 87% accuracy and an inspection speed of 22 FPS. However, at present, there is still little research on real-time surface defect inspection based on semi-supervised and weakly supervised learning.
- (4) **Unsupervised learning.** Unsupervised defect inspection methods only use defect-free samples to train the model, and then the model has the ability to reconstruct defect-free samples. When there is a defect in the input image, the model will reconstruct the corresponding defect-free image, and realize the inspection of the defect through the reconstruction error between the input image and the reconstructed image. Unsupervised learning methods solve the difficulty of collecting defect samples and labeling samples in industry. However, compared with supervised learning methods, the accuracy of unsupervised methods is lower, especially when the image background is complex. Defect inspection methods based on unsupervised learning are rarely used in practical industrial scenarios, and further research is needed. Yang et al. (2019b) proposed a multiscale feature clustering-based fully convolutional autoencoder, which can effectively and accurately detect multiple types of texture defects with a small number of defect-free samples. The autoencoder consists of a feature extraction network, a multi-scale fully convolutional autoencoder network, and a result fusion module. The feature extraction network is used to extract features with different receptive fields from the input image, which are then fed into different fully convolutional autoencoders to reconstruct the image at different scales. The result fusion module fuses the residual images of different scales obtained from the input image and the reconstructed image, and outputs the final defect inspection result. Through experiments on multiple textured

surface datasets, the proposed autoencoder achieves 92% precision, and it only takes 82ms to detect an image with the resolution of 1920×1080 . Reconstructing defect images using autoencoders (Bergmann et al. 2018; Tian and Li 2019; Youkachen et al. 2019) is the most commonly used image reconstruction method. However, the defect-free image reconstructed by the autoencoder is more similar to the input defect image than the equivalent defect-free image (Bhattad et al. 2018). To solve the problem of autoencoders “peeping” at the input image, Oz et al. (2021) proposed a nested autoencoder architecture. First, for a defect image, a one-dimensional vector is obtained through the encoder, denoted as a defect vector. Then, a defect-free vector is predicted through an autoencoder composed of fully connected layers, and a defect image and a defect-free image are generated according to the defect vector and the defect-free vector, respectively. Finally, an error map is obtained based on the newly generated two images for defect segmentation. The approach achieves 78% mIoU at 88ms per image on the MVTecAD dataset.

5.2 Model

Generally, the most important metric to measure the performance of deep learning models is accuracy. However, in industry, model inference speed is also an important metric for evaluating model performance. For defect inspection methods based on deep learning, high accuracy can effectively detect defects in products, which is conducive to product quality control, and real-time inspection can detect and screen defective products in time, which is conducive to improving production efficiency. The powerful feature extraction capability of the deep learning model is realized by its complex network structure and network depth, but more complex network structure and deeper network depth decrease the inference speed of the model. High accuracy and real-time inspection are the goals of industrial surface defect inspection. Therefore, a trade-off between the accuracy and speed of the deep learning model is required, that is, a trade-off between feature extraction capability and model scale. The following are commonly used methods to improve the model feature extraction capability or reduce the model scale.

- (1) Lightweight backbone network. Currently, representative lightweight classification networks include SqueezeNet, MobileNet, ShuffleNet (Zhang et al. 2018), etc. These classification networks have the advantages of strong feature extraction capability and small network scale. And these networks have been applied to various computer vision tasks, and the performance of these networks has been verified. These lightweight classification networks are also used as backbone networks for surface defect inspection models, and show high performance. Fu et al. (2019) and Yang et al. (2020c) used SqueezeNet as the backbone network of the surface defect inspection model, and used it for steel strip surface defect inspection and USB connector defect inspection, respectively. Zhang et al. (2020b) and Li et al. (2018b) used MobileNet for vehicle body paint surface defect inspection and container Sealing surface defect inspection, respectively.
- (2) Feature pyramid network. Feature pyramid network (FPN) obtains features that contain both rich semantic information and accurate location information by fusing the shallow features and deep features in the backbone network. FPN can effectively solve the problem of insufficient model feature extraction ability. Besides, PANet, balanced feature pyramid (BFP) (Pang et al. 2019), bidirectional feature pyramid network (BiFPN) (Tan

et al. 2020) etc. are also commonly used. Huang et al. (2021a) introduced PANet in the plastic relay surface defect inspection and improved PANet based on the self-attention mechanism. Hao et al. (2021) used BFP for steel strip surface defect inspection. BFP enhances features of different resolutions by balancing semantic features. Medak et al. (2022) introduced BiFPN in the defect inspection model, and BiFPN learns the importance of features at different scales by learning weights.

- (3) Attention mechanism. Introducing attention mechanism into the surface defect inspection model can make the model focus more on the defect area in the image, and extract features more effectively. Currently, the attention mechanisms that are widely used in surface defect inspection mainly include spatial attention, channel attention and self-attention. Dong et al. (2021) combined spatial attention mechanism with channel attention mechanism in pavement surface defect inspection. Chen et al. (2021b) added different weights to features at different positions in the feature map through the self-attention mechanism, which increased the influence of important features on the final inspection results.
- (4) Anchor-Free based object detection model. In particular, the current object detection models widely used in surface defect detection, such as YOLOv3, YOLOv4, Faster RCNN, etc., are based on Anchor. For the Anchor-based defect detection model, multiple bounding box regressions are performed on each pixel of the feature map according to multiple preset Anchors, and consume a lot of memory resources and computing time. Anchor-Free-based defect detection methods (Yu et al. 2021; He et al. 2022) overcome the shortcomings of Anchor-based defect detection methods and can improve the speed of surface defect detection.

5.3 Application

It is not enough for surface defect inspection methods to show high performance on public datasets, but more importantly, in industrial scenarios. During model training and testing, the experimental environment is vastly different from the industrial environment, which often results in surface defect inspection methods not performing as expected in industrial scenarios. Unlike general computer vision tasks, the industrial environment is also a factor that needs to be considered when designing surface defect inspection methods. The complex industrial environment brings great challenges to surface defect inspection, and there are still problems that need to be solved urgently. This section presents the challenge faced by surface defect inspection methods from the application perspective.

- (1) Low quality image. In public datasets, image samples are preprocessed and selected to facilitate training and testing of the surface defect inspection model. However, in industry, the images acquired by image acquisition equipment often have various noises, motion blur, uneven illumination, etc., which reduce the performance of surface defect inspection methods. To reduce the interference of these factors, methods of enhancing contrast and filtering noise are often used to improve image quality. Commonly used methods for enhancing image contrast include histogram equalization (Jun et al. 2021) and grayscale transformation (Wei et al. 2019a, 2019b), while methods for filtering noise include median filtering (Wei et al. 2019b) and Gaussian filtering (Zhang et al. 2021). Median filtering can not only reduce the interference of salt and pepper noise on the image, but also avoid blurring the edges of objects, while Gaussian filtering can sharpen the image and enhance the details in the image.

- (2) High-resolution image. Both training and testing of surface defect inspection models are based on low-resolution images, but in industry, the images acquired by image acquisition equipment are all high-resolution. Directly inputting the acquired images into the defect inspection model will seriously slow down the inspection speed and make real-time inspection impossible. To adapt to the input size of the model, down-sampling the image will cause the loss of defect information, or even the disappearance of small defects. The most common method is to crop high-resolution images and feed them separately into surface defect inspection models. However, cropping the image may result in incomplete defects and affect inspection performance. In particular, Dlamini et al. (2022) used the Distance Matching Function (Oh et al. 1999) to decompose the image. At present, there is still a lack of effective methods for detecting high-resolution images, which is also an urgent problem to be solved in the surface defect inspection task.
- (3) New types of defects. Due to the complex industrial environment and the uncertainty of defect occurrence, new types of surface defects may arise. Surface defect inspection models trained based on existing types of defects are difficult to detect new types of defects effectively, which reduces the performance of surface defect inspection models. Therefore, the defect inspection model needs to be able to learn new types of defects and has the ability to detect new types of defects.

This section discusses the challenges faced by deep learning-based surface defect inspection methods from the perspective of the dataset, model, and application, and introduces commonly used solutions. Especially in the application, there are still problems that need to be solved urgently. Table 10 summarizes the challenges faced by deep learning-based methods and commonly used solutions.

6 Conclusion and future trends

Surface defect inspection methods based on deep learning can accurately and quickly perform non-destructive inspection for industrial products. From the perspective of real-time inspection, this paper reviews the defect inspection methods for metal strip surface, metal product surface, textured surface, other types of surfaces, and general surface. For each type of surface defect inspection method, this paper reviews the current research in more detail from three levels: defect classification, defect detection, and defect segmentation. Then, the paper introduces commonly used metrics for evaluating surface defect inspection models and publicly available defect datasets. Furthermore, this paper discusses the challenges faced by deep learning-based surface defect inspection methods from the perspective of the dataset, model and application. However, the commonly used solutions cannot completely overcome the challenges faced by surface defect inspection, and more new technologies are needed.

For the challenge of lacking training samples, domain-adaptive transfer learning and federated learning are expected to be new solutions. Domain-adaptive transfer learning can project the source and target domains into the same feature space, enabling the model to learn features across domains. Domain-adaptation can solve the problem of the distribution difference between training samples and test samples, and can reduce the excessive dependence of deep learning models on the distribution of training samples. Through domain-adaptation, the model can learn knowledge from a large number of labeled source

Table 10 Challenges faced by deep learning-based methods and commonly used solutions

Perspective	Challenge	Solution		
Dataset	In the industrial production process, collecting a certain number of samples is a difficult and time-consuming process, and labeling the collected samples is also a labor-intensive process. The number of defect samples is far from enough for deep learning method.		Data augmentation	
	High-accuracy inspection is conducive to product quality control, while real-time inspection is conducive to improving production efficiency. To achieve high accuracy and real-time inspection, a trade-off between the accuracy and speed of the deep learning model is required.	Increase the number of training samples	Transfer learning Semi-supervised and weakly supervised learning Unsupervised learning Feature pyramid network	
		Reduce dependence on sample labels		
Model	Low quality image High-resolution image New type of defect	Improve the feature extraction capability	Attention mechanism Lightweight backbone network Anchor-free object detection	
		Reduce the model scale		
		Contrast enhancement and noise filtering. Image cropping and distance matching function Ability to learn new types of defects		
Application				

domain samples and transfer the knowledge to the target task. Applying domain-adaptive transfer learning to surface defect inspection can alleviate the lack of training samples to some extent. Federated learning is an emerging distributed machine learning method that can share data while protecting data privacy. The core of federated learning is distributed model training on multiple data sources. There is no data interaction between multiple data sources, only the interaction of model intermediate parameters, so as to realize “data is available but not visible”. In surface defect inspection, due to commercial confidentiality, factories seldom disclose datasets, which leads to limited public surface defect datasets. Federated learning can make it possible to train surface defect inspection models on datasets from multiple factories under the premise of ensuring data privacy.

For the challenge of balancing the model accuracy and speed, inspection methods based on 3D point cloud and neural network verification that have emerged in recent years are expected to be applied to industrial surface defect inspection. 3D Point cloud data is the point collection of the target in three-dimensional space, which can accurately represent the surface state of the target. Compared with 2D images, 3D point cloud data contains three-dimensional feature information of defects, such as the height and depth of defects, and is not easily affected by uneven illumination. 3D point cloud data contains more defect-related information, and surface defect inspection methods based on point cloud data can extract more defect features and achieve higher accuracy. Surface defect inspection based on 3D point cloud or the combination of 2D images and 3D point cloud data may become the future trend. For deep learning, neural networks are often regarded as “black-box” models that lack interpretability. Neural network verification, as an emerging field, can verify the properties of neural networks. For example, in the robustness verification, it is verified that no matter what attack method is used on an image, as long as the perturbation does not exceed a certain threshold, the attack will not succeed. Neural network verification provides interpretability for neural network behavior. Based on neural network verification, a surface defect inspection algorithm with higher accuracy and faster speed is expected to be proposed.

For the challenge of the application in industry, there is still a lack of new approaches to address low quality images, high-resolution images, and new types of defects. For low quality images, the most commonly used methods are still the traditional methods of enhancing contrast and filtering noise. For high-resolution images, cropping-based methods are still the most commonly used. In the future, new and more effective image processing methods are expected to be proposed. For new types of defects, the surface defect inspection method needs to be able to learn new samples in the practical application, and continuously improve the inspection ability based on new samples.

In this paper, current defect inspection methods for different types of surfaces in industry are reviewed from the perspective of real-time. Based on this review, relevant researchers can obtain the research status of real-time surface defect inspection. In future studies, this review can help researchers choose different real-time defect inspection methods according to different surface characteristics. Furthermore, this review summarizes the current challenges in real-time defect inspection, which provide future research directions. Finally, this review presents many future research trends that can be taken into consideration by researchers in this area. This review is expected to provide an effective reference for relevant researchers.

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